

Homework Assignment 4

1. An ideal pump pressurizes hydrogen ($SG = 0.071$) from 140 kPa to 4 MPa. The flow rate is 45 kg/s. What is the power (sign and magnitude) of the pump? Is the power delivered or absorbed by the pump? {Ans.: $|\mathcal{P}| = 2.45$ MW}
2. An ideal turbine is used to drive the pump of the previous problem. Given that the entry and exit stagnation temperatures are 1128 K and 818 K, respectively, calculate the required mass flow rate of the working fluid ($\gamma = 1.3$, $\bar{M} = 10.5$). Also, find the turbine expansion ratio. {Ans.: $\dot{m}_t = 2.3$ kg/s}
3. An isentropic, c-d nozzle has area ratio 10. Combustion gases enter it at stagnation pressure of 6 MPa. Assuming supersonic exit flow in sea-level operation ($p_a = 0.1$ MPa), calculate the nozzle exit pressure. Is the exit jet under-, fully- or over-expanded? Calculate the angle of the shock formed at the exit plane. $\gamma = 1.25$. (Note: The area ratio determines the Mach number.) {Ans.: $\sigma = 20.74^\circ$ }